1 A student is investigating the behaviour of a capacitor-resistor circuit as shown in Fig. 1.1.

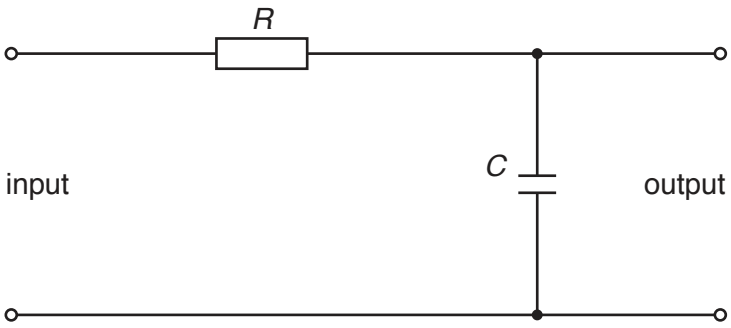


Fig. 1.1

A neon lamp flashes on and off when it is connected across the capacitor with a potential difference V_F across the lamp of approximately 90 V.

The student has a number of unmarked resistors.

It is suggested that the period T of the flashes of the lamp is related to the resistance R of the resistor by the expression

$$T = RCK$$

where C is the capacitance of the capacitor and K is a constant.

The constant K is given by

$$K = \ln\left(\frac{V_i - V_L}{V_i - V_F}\right)$$

where V_i is the potential difference across the input, V_F is the potential difference required to make the lamp flash and V_L is a constant.

Design a laboratory experiment to test the relationship between T and R.
Explain how your results could be used to determine a value for K and V_L.

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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